

Abstract

This paper studies simple fractions, probabilities and comparisons. We show that:

Within any interval $2n + 1$, that is between any two consecutive square numbers of n and $n+1$, number of primes $\geq 2 + \frac{1}{q}$, where q is the nearest prime $\leq n$.

Introduction:

Using Methods of Sieve of Eratosthenes,

Probability of prime numbers are reduced after subtracting the multiples of 2 and more.

For independent events,

New probability of getting primes = $P(A) - P(A) \times P(B)$, where $P(A)$ is the old probability and $P(B)$ is the probability of getting multiples of prime numbers.

From the multiples of prime number 2, the probability of getting primes after 4 is:

$$1 - \frac{1}{2} = \frac{1}{2}$$

Means from 4 to 8, one prime number is found in every 2 numbers.

From the multiples of prime number 3, the probability of getting primes after 9 is:

$$\frac{1}{2} - \frac{1}{6} = \frac{1}{3}$$

Means from 9 to 24, one prime number is found in every 3 numbers.

Here, we are subtracting the probability of getting the multiples of the prime numbers.

Let the probability to get the multiple of a prime number(p) be $\frac{1}{p}$

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Put $p = 3$, because until prime number 3 the difference of consecutive prime number is 1. So, new probability will be $\frac{1}{p+1}$ Let, initial probability of primes $P(A)$ be $\frac{1}{p} = \frac{1}{3}$. (We are taking initial probability as $\frac{1}{3}$ because from prime number 3, the consecutive primes will have minimum difference of 2(for calculations))

We know, the probability of primes decreases as we subtract the multiples of 5 and more.

Since, the prime will be in minimum difference of 2 from prime-3. So, we use $(p+2)$ for the next prime numbers.

New probability = $\frac{1}{p} - \frac{1}{p(p+2)}$equation(1)

To prove: New probability is greater than $\frac{1}{(p+2)}$.

We know, $\frac{1}{n} - \frac{1}{n+2} > \frac{1}{n(n+2)}$

where n is natural number (specially) greater than 2, starts from 3 as odd number.

Or, $\frac{1}{n} - \frac{1}{n(n+2)} > \frac{1}{n+2}$

Or, $\frac{1}{n+2} < \frac{1}{n} - \frac{1}{n(n+2)}$

Comparing with equation (1), replacing n with p, since both are odd number greater than 2, with minimum difference of 2 with consecutive term. Also, n may be a prime(p) or greater than prime(p) and similar for n+2 and p+2.

$$\frac{1}{p+2} < \frac{1}{p} - \frac{1}{p(p+2)}$$

Since RHS is equal to new probability.

Comparing RHS with new probability,

New probability is always greater than $\frac{1}{p+2}$

Conclusion: the probability of getting a prime number is strictly greater than $\frac{1}{p+2}$, where p+2 is the latest prime number so that we subtract their multiples from their square numbers.

Also, $\frac{1}{p+2}$ will be new $\frac{1}{p}$ for next cases, the situation for “new probability” remains the same, as new probability is more than $\frac{1}{p}$.

$$\text{Given: } (n+1)^2 - n^2 = 2n+1$$

Let the difference between two consecutive squares be 2n+1 where n is one interval, another n is another interval.

Let (q) be a prime number nearest to number (n). So, $n \geq q$.

The interval between n^2 and $(n+1)^2 \geq 2q+1$

We take a prime number q because, from the square of q that is q^2 or $\leq n^2$, probability of getting prime numbers will be reduced.

From above conclusion: The probability of getting a prime number is strictly greater than $\frac{1}{(p+2)}$.

Let $(p+2) = q$, as from there we will get new probability of getting primes and q is the nearest prime number to n.

New probability of getting primes (P) $\geq \frac{1}{q}$

Interval(Z) from n^2 to $(n+1)^2 = 2q+1$

Note: $n = n$, $p = p$, $q = q$.

So, number of primes:

$$= P \times Z$$

$$\geq \frac{1}{q} \times (2q+1)$$

$$\geq \frac{2q+1}{q}$$

$$\geq 2 + \frac{1}{q}$$

Hence, within any interval $2n + 1$ number of prime $\geq 2 + \frac{1}{q}$, where q is the nearest prime lesser or equal to n .

Conclusion: For any positive integer n , there exists at least two prime number q such that

$$n^2 < q < (n + 1)^2.$$